

Case Study 1 – Employee Salaries

Introduction

Statistical analysis is an essential tool used by organizations to understand employee salary distribution and make informed decisions regarding compensation, budgeting, promotions, and workforce planning. Measures of central tendency such as the **mean, median, and mode** help identify the typical salary, while measures of dispersion like the **range, variance, and standard deviation** indicate how spread out the salaries are. Detecting outliers is equally important because extremely high or low salaries can significantly affect statistical results and business decisions.

In this case study, the annual salaries (in lakhs) of nine employees are analyzed using descriptive statistical techniques.

Given Data

Employee Salaries (in lakhs):

4, 5, 5, 6, 7, 8, 8, 9, 30

Number of observations (**n**) = **9**

1. Find the Mean, Median, and Mode

Mean (Average)

The arithmetic mean is calculated using the formula:

$$\begin{bmatrix} \text{Mean} = \Sigma X / n \end{bmatrix}$$

Where:

- ΣX = Sum of all salaries
- n = Number of employees

Calculation

Sum of salaries

$$= 4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30$$

$$= 82$$

Therefore,

$$\begin{bmatrix} \text{Mean} = 82 \div 9 = 9.11 \end{bmatrix}$$

Mean = 9.11 lakhs

Median

The median is the middle value when data is arranged in ascending order.

Ordered data:

4, 5, 5, 6, 7, 8, 8, 9, 30

Since there are 9 observations,

Median position

$$\begin{bmatrix} \text{Median Position} = (9 + 1) \div 2 = 5\text{th observation} \end{bmatrix}$$

The 5th value is 7.

Median = 7 lakhs

Mode

Mode is the value that appears most frequently.

Frequency table

Salary	Frequency
4	1
5	2
6	1
7	1
8	2
9	1
30	1

Both **5** and **8** appear twice.

Mode = 5 lakhs and 8 lakhs

Hence, the data is **bimodal**.

2. Find the Range

Range measures the spread between the maximum and minimum values.

Formula

[
Range = Maximum Value – Minimum Value
]

Maximum salary = **30**

Minimum salary = **4**

Therefore,

[
Range=30-4=26
]

Range = 26 lakhs

Interpretation

A range of 26 lakhs indicates a large difference between the highest-paid employee and the lowest-paid employee. This suggests significant salary variation within the company.

3. Calculate the Variance and Standard Deviation

Variance measures how far each salary deviates from the mean.

Standard deviation is the square root of variance.

Mean = **9.11**

Salary	X-Mean	(X-Mean) ²
4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.67
7	-2.11	4.45
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.39

Sum of squared deviations

≈ **512.88**

Variance

[
Variance = $512.88 \div 9 = 56.99$
]

= **56.99**

Variance = 56.99

Standard Deviation

[
Standard Deviation = $\sqrt{56.99} = 7.55$
]

= **7.55**

Standard Deviation = 7.55 lakhs

Interpretation

A standard deviation of **7.55 lakhs** shows that salaries are widely spread around the average. This large deviation is mainly due to one employee earning **30 lakhs**, which is much higher than the others.

4. Identify the Outlier

An outlier is a value that differs significantly from the rest of the observations.

The salary data mostly lies between **4 and 9 lakhs**, while one employee earns **30 lakhs**.

Since **30 lakhs** is far away from all other salary values, it is considered an **outlier**.

Outlier = 30 lakhs

Why is it an outlier?

- Much larger than the remaining salaries.
- Increases the average considerably.
- Causes a large standard deviation.
- Indicates exceptional pay, possibly for a senior executive or company director.

5. Which Measure of Central Tendency Best Represents the Typical Salary? Explain

The three measures of central tendency are:

- Mean = **9.11 lakhs**
- Median = **7 lakhs**
- Mode = **5 lakhs and 8 lakhs**

Among these, the **median** best represents the typical employee salary.

Reasons

1. The salary of **30 lakhs** is an extreme value.
2. The mean is pulled upward because of this outlier.

3. The median is not affected by extremely large or small values.
4. Most employees earn between **4 and 9 lakhs**, making **7 lakhs** a better representation of the typical salary.
5. Businesses commonly use the median salary when reporting employee earnings because it reflects the middle employee rather than being distorted by unusually high salaries.

Overall Interpretation

The analysis shows that the company has one exceptionally high-paid employee compared to the rest of the workforce. Although the average salary is **9.11 lakhs**, this value is inflated by the outlier of **30 lakhs**. The median salary of **7 lakhs** provides a more accurate picture of what a typical employee earns. The high range and standard deviation indicate considerable salary variation within the organization.

Conclusion

This case study demonstrates how descriptive statistics help analyze employee salary data effectively. The **mean (9.11 lakhs)**, **median (7 lakhs)**, and **bimodal values (5 and 8 lakhs)** describe the central tendency, while the **range (26 lakhs)**, **variance (56.99)**, and **standard deviation (7.55 lakhs)** explain salary variability. The **30-lakh salary** is identified as an outlier that significantly influences the mean. Therefore, the **median** is the most suitable measure for representing the typical salary because it is resistant to extreme values and better reflects the earnings of most employees. This analysis supports fair salary evaluation and informed organizational decision-making.

Case Study 2 – Website Response Time

Introduction

Website response time is one of the most important performance indicators in web applications. It measures the time taken by a website or server to respond to a user's request. A fast response time improves user satisfaction, enhances search engine rankings, and increases business revenue. Data scientists and

system administrators regularly analyze response time data to identify performance bottlenecks and optimize system efficiency.

In this case study, response times (in milliseconds) for ten website requests are analyzed using statistical methods. Measures of central tendency, dispersion, and skewness help understand the website's overall performance and determine whether unusual delays exist.

Given Data

Response Times (milliseconds):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Number of observations (**n**) = **10**

1. Calculate the Mean and Median

Mean (Average)

The mean is calculated using:

$$\begin{aligned} &[\\ \text{Mean} &= \Sigma X / n \\ &] \end{aligned}$$

Step 1: Find the Sum

$$\begin{aligned} &120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 \\ &= \mathbf{1505} \end{aligned}$$

Step 2: Divide by Number of Observations

$$\begin{aligned} &[\\ \text{Mean} &= 1505 \div 10 = 150.5 \\ &] \end{aligned}$$

Mean = 150.5 milliseconds

Interpretation

The average response time is **150.5 ms**. However, one unusually high value (300 ms) increases the average significantly.

Median

Since there are **10 observations (even number)**,

Median is the average of the 5th and 6th observations.

Ordered Data:

120, 125, 128, 130, **132, 135**, 140, 145, 150, 300

Median

[
Median = $(132 + 135) \div 2$
]

[
Median = $267 \div 2 = 133.5$
]

Median = 133.5 milliseconds

Interpretation

The median represents the middle response time and is not influenced by the unusually large response time of 300 ms.

2. Find the Range

The range measures the difference between the maximum and minimum values.

Formula

[
Range=Maximum-Minimum
]

Maximum = **300 ms**

Minimum = **120 ms**

Therefore,

[
Range = $300 - 120 = 180$
]

Range = 180 milliseconds

Interpretation

A range of **180 ms** indicates that there is a large variation between the fastest and slowest website responses. This suggests that while the website usually performs well, some requests experience significantly longer delays.

3. Calculate the Variance and Standard Deviation

Mean = **150.5**

Response Time	X – Mean	(X – Mean) ²
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22350.25

Total of squared deviations

= **25581.50**

Variance

[

Variance = $25581.5 \div 10 = 2558.15$

]

= **2558.15**

Variance = 2558.15

Standard Deviation

[

Standard Deviation = $\sqrt{2558.15} = 50.58$

]

≈ **50.58 milliseconds**

Standard Deviation = 50.58 milliseconds

Interpretation

The standard deviation is relatively high because of the **300 ms** response time. Most response times lie between **120 and 150 ms**, but the unusually slow request greatly increases the variability.

4. Determine Whether the Data is Right-Skewed or Left-Skewed

Skewness indicates whether data is stretched more on one side of the distribution.

Observations

- Most response times are clustered between **120 and 150 ms**.
- One value (**300 ms**) is much larger than the others.
- Mean (**150.5 ms**) is greater than the median (**133.5 ms**).

When the mean is greater than the median due to a few high values, the distribution is **positively skewed (right-skewed)**.

Answer

The data is **right-skewed**.

Explanation

The single response time of **300 ms** creates a long tail on the right side of the distribution. This indicates that although the website generally responds quickly, occasional requests experience unusually long delays.

5. Why is the Median More Useful Than the Mean?

Both the mean and median measure the center of the data, but they behave differently in the presence of outliers.

Mean

- Uses every observation.
- Easily affected by extreme values.

- Gives an average that may not represent the majority of users.

Median

- Represents the middle value.
- Not affected by outliers.
- Better reflects the response time experienced by most users.

In this dataset

Mean = **150.5 ms**

Median = **133.5 ms**

The mean suggests the website is slower than it actually is because of the **300 ms** outlier. The median shows that half the requests are completed in **133.5 ms or less**, making it a more realistic measure of typical website performance.

Overall Interpretation

The statistical analysis reveals that the website generally performs efficiently, with most response times ranging from **120 to 150 milliseconds**. However, one unusually slow response of **300 milliseconds** increases the mean, variance, standard deviation, and overall range. This indicates occasional performance issues, such as temporary server overload, network congestion, or database delays.

The distribution is **positively skewed**, meaning that most users experience fast response times, while a small number of users encounter significantly slower responses. For evaluating the typical user experience, the **median** provides a more reliable measure than the mean.

From a business perspective, identifying and reducing these occasional high response times can improve customer satisfaction, website reliability, and system efficiency.

Conclusion

This case study demonstrates the importance of descriptive statistics in analyzing website performance. The **mean response time is 150.5 ms**, while

the **median is 133.5 ms**, indicating that the mean is influenced by the unusually slow response of **300 ms**. The **range (180 ms)**, **variance (2558.15)**, and **standard deviation (50.58 ms)** show considerable variability due to this outlier. The data is **right-skewed**, highlighting that a few slow responses can significantly affect overall performance metrics. Therefore, the **median** is the most appropriate measure for representing the typical website response time because it accurately reflects the experience of most users without being distorted by extreme values.

Case Study 3 – Student Performance

Introduction

Statistical analysis is widely used in education to evaluate students' academic performance. By using descriptive statistics such as quartiles, interquartile range (IQR), mean, median, and outlier detection, educators can understand the distribution of marks and identify exceptional performances. Quartiles divide the data into four equal parts, while the IQR measures the spread of the middle 50% of the data. The **$1.5 \times \text{IQR}$ rule** is commonly used to detect unusually high or low scores (outliers).

In this case study, the marks of 12 students in a Data Science test are analyzed to determine quartiles, identify outliers, calculate measures of central tendency, and study the effect of an outlier on the mean.

Given Data

Marks obtained by 12 students:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Number of observations (**n**) = **12**

1. Find Q1, Q2, and Q3

Quartiles divide the ordered data into four equal parts.

Step 1: Arrange the Data

The data is already in ascending order:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

First Quartile (Q1)

The lower half consists of:

42, 45, 48, 50, 52, 52

Since there are six values, Q1 is the average of the 3rd and 4th values.

$$\begin{aligned} &[\\ Q_1 &= (48 + 50) \div 2 = 49 \\ &] \end{aligned}$$

$$\mathbf{Q1 = 49}$$

Second Quartile (Q2 / Median)

Since there are 12 observations,

Median is the average of the 6th and 7th values.

6th value = 52

7th value = 55

$$\begin{aligned} &[\\ Q_2 &= (52 + 55) \div 2 = 53.5 \\ &] \end{aligned}$$

$$\mathbf{Q2 (Median) = 53.5}$$

Third Quartile (Q3)

Upper half:

55, 58, 60, 62, 65, 95

Q3 is the average of the 3rd and 4th values.

$$\begin{aligned} &[\\ Q_3 &= (60 + 62) \div 2 = 61 \\ &] \end{aligned}$$

$$\mathbf{Q3 = 61}$$

Summary of Quartiles

Quartile	Value
Q1	49
Q2 (Median)	53.5
Q3	61

2. Calculate the Interquartile Range (IQR)

The **Interquartile Range (IQR)** measures the spread of the middle 50% of the data.

Formula

$$\begin{aligned} &[\\ \text{IQR} &= Q_3 - Q_1 \\ &] \end{aligned}$$

Substituting the values:

$$\begin{aligned} &[\\ \text{IQR} &= 61 - 49 = 12 \\ &] \end{aligned}$$

$$\text{IQR} = 12$$

Interpretation

An IQR of **12 marks** means that the middle 50% of students scored within a range of 12 marks, indicating that most students performed similarly.

3. Use the $1.5 \times \text{IQR}$ Rule to Identify Outliers

The $1.5 \times \text{IQR}$ rule is used to detect unusually high or low values.

Step 1: Calculate $1.5 \times \text{IQR}$

$$\begin{aligned} &[\\ 1.5 \times 12 &= 18 \\ &] \end{aligned}$$

Step 2: Find the Lower and Upper Limits

Lower Limit

$$[$$

$$Q_1 - 18 = 49 - 18 = 31$$

$$]$$

Upper Limit

$$[$$

$$Q_3 + 18 = 61 + 18 = 79$$

$$]$$

Step 3: Compare the Data

Any value:

- Below **31**
- Above **79**

is considered an outlier.

The only value above **79** is:

95

Outlier = 95

Explanation

The score of **95** is much higher than the rest of the class, whose marks range mostly between **42 and 65**. Therefore, **95** is considered an outlier.

4. Find the Mean and Median

Mean

Formula:

$$[$$

$$\text{Mean} = \Sigma X / n$$

$$]$$

Sum of Marks

$$42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95$$

$$= \mathbf{684}$$

[
Mean = $684 \div 12 = 57$
]

Mean = 57 marks

Median

Median is already calculated as:

Median = 53.5 marks

Comparison

Measure Value

Mean **57**

Median **53.5**

The mean is higher than the median because of the exceptionally high score of **95**.

5. Explain How the Outlier Affects the Mean

An **outlier** is a value that differs significantly from the rest of the observations.

In this dataset:

- Most students scored between **42 and 65**.
- One student scored **95**, which is much higher than the other marks.

Effect on the Mean

The mean uses **every value** in the dataset. Therefore, the score of **95** increases the total sum of marks, causing the average to become **57**, which is higher than the marks obtained by most students.

Effect on the Median

The median depends only on the middle values and is **not influenced** by extreme observations. Even though one student scored **95**, the median remains **53.5**, giving a better representation of the typical student's performance.

Practical Importance

- If teachers use only the **mean**, they may believe the class performed better than it actually did.
- The **median** provides a more realistic picture of the overall class performance.
- Detecting outliers helps educators identify exceptionally high-performing students without allowing their scores to distort the class average.

Overall Interpretation

The analysis shows that the majority of students performed consistently, with marks concentrated between **42 and 65**. The quartiles (**Q1 = 49, Q2 = 53.5, and Q3 = 61**) indicate that the middle 50% of students scored within a relatively narrow range. The **IQR of 12** confirms moderate variation among most students.

Using the **$1.5 \times \text{IQR}$ rule**, the score of **95** is identified as an outlier because it lies above the upper limit of **79**. This exceptionally high score increases the **mean (57)** but has little effect on the **median (53.5)**. Therefore, the median is a more reliable measure of the typical student's performance.

Such statistical analysis enables teachers and educational institutions to evaluate academic performance more accurately, recognize outstanding students, and make informed decisions regarding teaching strategies and student assessment.

Conclusion

This case study demonstrates the application of descriptive statistics in analyzing student performance. The quartiles (**Q1 = 49, Q2 = 53.5, Q3 = 61**) provide insight into the distribution of marks, while the **IQR of 12** measures the spread of the middle half of the data. Using the **$1.5 \times \text{IQR}$ rule**, the score of **95** is identified as an outlier. The **mean (57)** is influenced by this exceptionally high mark, whereas the **median (53.5)** remains unaffected and better represents

the performance of the majority of students. These statistical tools help educators interpret examination results accurately and support effective academic decision-making.

Case Study 4 – E-Commerce Orders

Introduction

In the e-commerce industry, analyzing daily order data is essential for understanding customer demand, inventory planning, staffing, and business growth. Statistical measures such as the **mean, median, mode, variance, standard deviation, and coefficient of variation** help businesses evaluate the consistency and variability of customer orders. These measures allow managers to identify trends, predict future demand, and improve operational efficiency.

In this case study, an online store records the number of orders received over ten consecutive days. The data is analyzed using descriptive statistics to understand the overall order pattern and the variability in customer demand.

Given Data

Number of orders over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Number of observations (**n**) = **10**

1. Calculate the Mean, Median, and Mode

Mean (Average)

The mean is calculated using the formula:

$$\begin{aligned} &[\\ \text{Mean} &= \Sigma X / n \\ &] \end{aligned}$$

Step 1: Calculate the Sum

$$\begin{aligned} &18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 \\ &= \mathbf{250} \end{aligned}$$

Step 2: Divide by Number of Observations

[
Mean = $250 \div 10 = 25$
]

Mean = 25 orders

Interpretation

On average, the online store receives **25 orders per day**.

Median

Since there are **10 observations**, the median is the average of the 5th and 6th values.

Ordered data:

18, 20, 22, 22, **24, 25**, 26, 28, 30, 35

[
Median = $(24 + 25) \div 2 = 24.5$
]

Median = 24.5 orders

Interpretation

Half of the days recorded fewer than **24.5 orders**, while the other half recorded more.

Mode

The mode is the value occurring most frequently.

Frequency:

- 22 appears **2 times**
- All other values appear once.

Mode = 22 orders

Interpretation

The most common number of daily orders is **22**.

2. Calculate the Variance and Standard Deviation

Mean = 25

Orders	X – Mean	(X – Mean) ²
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

Sum of Squared Deviations

$$49 + 25 + 9 + 9 + 1 + 0 + 1 + 9 + 25 + 100$$

$$= 228$$

Variance

[

$$\text{Variance} = 228 \div 10 = 22.8$$

]

$$\text{Variance} = 22.8$$

Standard Deviation

[

$$\text{Standard Deviation} = \sqrt{22.8} = 4.77$$

]

$$\approx 4.77 \text{ orders}$$

$$\text{Standard Deviation} = 4.77 \text{ orders}$$

Interpretation

The daily orders usually vary by about **5 orders** from the average. This indicates that the order volume is fairly stable with only moderate fluctuations.

3. Calculate the Coefficient of Variation (CV)

The **Coefficient of Variation (CV)** measures the variability relative to the mean.

Formula

$$[\\ CV = (SD \div \text{Mean}) \times 100 \\]$$

Substituting the values:

$$[\\ CV = (4.77 \div 25) \times 100 \\]$$

$$[\\ CV = 19.08\% \\]$$

Coefficient of Variation = 19.08%

Interpretation

A CV of approximately **19%** indicates relatively low variability in daily orders. The order volume remains fairly consistent across the ten-day period.

4. What Does the Standard Deviation Tell the Company?

The **standard deviation** measures how much the daily orders differ from the average number of orders.

Interpretation of the Result

- Average daily orders = **25**
- Standard deviation = **4.77**

This means that most daily order counts fall within about **5 orders** of the average.

Business Importance

The standard deviation helps the company:

- Estimate daily demand fluctuations.
- Plan inventory efficiently.
- Schedule an appropriate number of employees.
- Reduce stock shortages or overstocking.
- Improve delivery and logistics planning.

Since the standard deviation is relatively small compared to the mean, the company can expect stable daily order volumes.

5. Would You Describe the Order Data as Highly or Moderately Variable?

To determine the variability, we examine the:

- Standard deviation = **4.77**
- Mean = **25**
- Coefficient of Variation = **19.08%**

Analysis

- A **low CV (below 20%)** generally indicates **low to moderate variability**.
- The daily orders increase gradually without extreme jumps.
- There are no unusually high or low order values.

Answer

The order data is **moderately variable**.

Explanation

The number of orders changes from day to day, but the variation is not large enough to be considered highly inconsistent. This indicates a steady business with predictable customer demand.

Overall Interpretation

The statistical analysis shows that the online store receives an average of **25 orders per day**, with the **median (24.5)** and **mode (22)** indicating that daily

orders are generally consistent. The **variance (22.8)** and **standard deviation (4.77)** reveal only moderate fluctuations in demand.

The **coefficient of variation (19.08%)** confirms that the variability is relatively low compared to the average number of orders. This suggests that customer demand is stable, allowing the company to manage inventory, staffing, and logistics more effectively.

From a business perspective, the analysis indicates that the company has predictable order patterns, enabling better operational planning and improved customer service.

Conclusion

This case study illustrates how descriptive statistics can be used to analyse e-commerce order data. The **mean (25 orders)**, **median (24.5 orders)**, and **mode (22 orders)** describe the central tendency of daily orders. The **variance (22.8)** and **standard deviation (4.77 orders)** indicate moderate variability, while the **coefficient of variation (19.08%)** confirms that the order volume is fairly consistent. Therefore, the company experiences **moderate variability** rather than high fluctuations, making demand forecasting and resource planning more reliable. Statistical analysis such as this supports informed decision-making and contributes to efficient business operations.

Case Study 5 – Comparing Two Data Science Models

Introduction

In machine learning and data science, evaluating the performance of predictive models is essential for selecting the most reliable algorithm. One common evaluation metric is the **prediction error**, which represents the difference between the predicted value and the actual value. A good model should have a **low average error** and **consistent performance** across different predictions.

Statistical measures such as the **mean**, **variance**, and **standard deviation** are widely used to compare prediction errors. The **mean** indicates the average prediction error, while the **variance** and **standard deviation** measure the consistency of the model. A model with a smaller standard deviation produces more stable predictions and is generally considered more reliable.

In this case study, two machine-learning models are compared using descriptive statistical analysis.

Given Data

Model A Prediction Errors:

2, 3, 3, 4, 4, 5, 5, 6

Model B Prediction Errors:

1, 1, 2, 4, 5, 7, 8, 9

Number of observations for each model (**n**) = **8**

1. Calculate the Mean Error for Both Models

Model A

Formula

$$\begin{bmatrix} \text{Mean} = \Sigma X / n \end{bmatrix}$$

Sum of Errors

$$2 + 3 + 3 + 4 + 4 + 5 + 5 + 6$$

$$= \mathbf{32}$$

$$\begin{bmatrix} \text{Mean} = 32 \div 8 = 4 \end{bmatrix}$$

$$\mathbf{\text{Mean Error (Model A)} = 4}$$

Model B

Sum of Errors

$$1 + 1 + 2 + 4 + 5 + 7 + 8 + 9$$

$$= \mathbf{37}$$

[
Mean = $37 \div 8 = 4.625$
]

Mean Error (Model B) = 4.625

Comparison

Model Mean Error

Model A **4.00**

Model B **4.625**

Interpretation

Model A has the **lower average prediction error**, indicating that, on average, its predictions are closer to the actual values than those of Model B.

2. Calculate the Variance for Both Models

Variance measures how much the prediction errors deviate from the mean.

Model A

Mean = **4**

Error	Error – Mean	(Error – Mean) ²
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

Sum of Squared Deviations

$4 + 1 + 1 + 0 + 0 + 1 + 1 + 4$

= 12

[
Variance = $12 \div 8 = 1.5$
]

Variance (Model A) = 1.5

Model B

Mean = **4.625**

Error	Error – Mean	(Error – Mean) ²
1	-3.625	13.14
1	-3.625	13.14
2	-2.625	6.89
4	-0.625	0.39
5	0.375	0.14
7	2.375	5.64
8	3.375	11.39
9	4.375	19.14

Sum of Squared Deviations

≈ 69.88

[
Variance = $69.88 \div 8 = 8.73$
]

Variance (Model B) = 8.73

Comparison

Model Variance

Model A **1.50**

Model B **8.73**

Interpretation

Model A has a much smaller variance, indicating that its prediction errors are more closely clustered around the mean.

3. Calculate the Standard Deviation for Both Models

Standard deviation is the square root of the variance.

Model A

$$\begin{aligned} &[\\ \text{Standard Deviation} &= \sqrt{1.5} = 1.22 \\ &] \end{aligned}$$

$$= 1.22$$

$$\text{Standard Deviation (Model A)} = 1.22$$

Model B

$$\begin{aligned} &[\\ \text{Standard Deviation} &= \sqrt{8.73} = 2.95 \\ &] \end{aligned}$$

$$\approx 2.95$$

$$\text{Standard Deviation (Model B)} = 2.95$$

Comparison

Model	Standard Deviation
-------	--------------------

Model A	1.22
---------	------

Model B	2.95
---------	------

Interpretation

Model A has a much lower standard deviation than Model B. This means its prediction errors vary less and are more consistent.

4. Which Model Has More Consistent Predictions?

Consistency refers to how closely the prediction errors are grouped around the average.

Model A

- Mean Error = 4
- Standard Deviation = 1.22
- Errors range from 2 to 6

Model B

- Mean Error = 4.625
- Standard Deviation = 2.95
- Errors range from 1 to 9

Analysis

Model A's prediction errors remain close to the mean, whereas Model B's errors fluctuate significantly. The lower variance and standard deviation of Model A indicate more reliable and stable predictions.

Answer

Model A has more consistent predictions because it has the lower variance (1.5) and standard deviation (1.22).

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better? Explain.

The mean error alone is **not sufficient** to determine which machine-learning model is better.

Reasons

1. Consistency Matters

A model may have a low average error but produce highly inconsistent predictions. Large fluctuations reduce the model's reliability.

2. Variability is Important

Variance and standard deviation show whether prediction errors remain stable across different observations.

3. Application Requirements

Some applications, such as medical diagnosis, financial forecasting, and autonomous vehicles, require highly consistent predictions rather than just a low average error.

4. Other Evaluation Metrics

Data scientists also consider several additional metrics, including:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Accuracy
- Precision
- Recall
- F1-Score
- R^2 Score

These metrics provide a more comprehensive evaluation of model performance.

Conclusion for This Dataset

In this dataset:

- **Model A** has the lower mean error (**4**) and much lower variability (**SD = 1.22**).
- **Model B** has a higher mean error (**4.625**) and greater variability (**SD = 2.95**).

Therefore, **Model A** is the better-performing model because it is both **more accurate on average and more consistent**.

Overall Interpretation

The comparison demonstrates the importance of using multiple statistical measures when evaluating machine-learning models. Although both models generate prediction errors, Model A performs better because its errors are smaller on average and much more consistent. The low variance and standard

deviation indicate that Model A produces stable predictions across different observations, while Model B shows greater fluctuations in prediction accuracy.

From a practical perspective, selecting a model based only on the mean error can be misleading. A complete evaluation should consider both **accuracy** and **consistency**, along with additional performance metrics relevant to the application. Such comprehensive analysis enables data scientists to choose models that provide reliable and dependable predictions in real-world scenarios.

Conclusion

This case study highlights the application of descriptive statistics in comparing machine-learning models. The analysis shows that **Model A** has a **mean error of 4**, **variance of 1.5**, and **standard deviation of 1.22**, whereas **Model B** has a **mean error of 4.625**, **variance of 8.73**, and **standard deviation of 2.95**. These results indicate that **Model A** produces lower average prediction errors and significantly more consistent predictions. However, model evaluation should not rely solely on the mean error; variability and other performance metrics must also be considered. By combining measures of central tendency and dispersion, organizations can make informed decisions when selecting the most effective machine-learning model for practical applications.